



SKYSHIELD

AI SECURITY ASSISTANT

Imagine Cup 



THE PROBLEM

- ▶ Alert Fatigue – Analysts are overwhelmed by excessive security alerts
- ▶ Poor Interpretability – Raw alerts lack clear context and explanation
- ▶ Difficult Prioritization – Requires experience to identify real threats
- ▶ High Cost Barrier – Enterprise SOC tools are expensive and inaccessible



YOUR SOLUTION: SKYSHIELD

- SkyShield is an AI-powered security assistant

It:

- ✓ Explains alerts in human language
 - ✓ Analyzes text + images
 - ✓ Assigns a risk score
 - ✓ Translates alerts in multiple languages
- Designed for:
 - ✓ SOC teams
 - ✓ Students
 - ✓ Cloud security learners

TECHNOLOGIES USED



Cloud & AI Services

- ▶ Azure AI Vision – Image-based security alert analysis (OCR & visual text extraction)
- ▶ Azure AI Language (Text Analytics) – Security alert text understanding and key insight extraction
- ▶ Azure Translator – Multi-language alert translation for global accessibility

Application & Backend

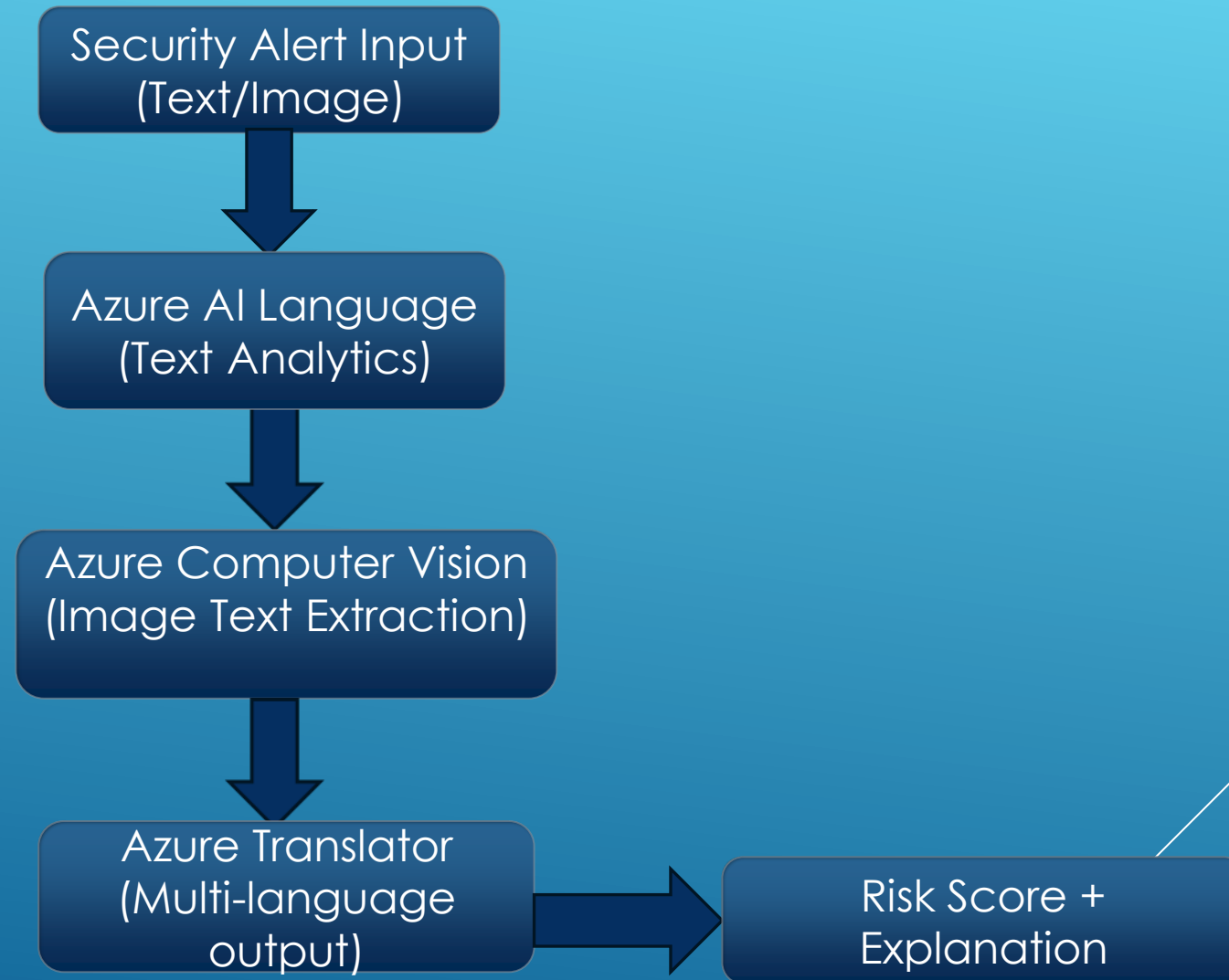
- ▶ Python – Core logic, AI service orchestration, and risk scoring engine
- ▶ Streamlit – Interactive MVP web interface for alert analysis

▶ Data & Visualization

- ▶ Pandas – Security alert data processing and structuring
- ▶ Plotly – Risk score visualization and alert analytics dashboards



VISUAL DIAGRAM





WHAT MAKES SKYSHIELD UNIQUE

- ▶ AI-Powered End-to-End Analysis – **Handles raw text and image-based alerts seamlessly**
- ▶ Multi-Language Support – **Alerts can be translated instantly for global accessibility**
- ▶ Risk Scoring & Prioritization – **Highlights critical threats for fast decision-making**
- ▶ Student & Junior-Friendly – **Cost-effective alternative to enterprise SOC tools**
- ▶ Solo-Developed MVP – **Lightweight, interactive, and easy-to-use via Streamlit**



FUTURE SCOPE

- Advanced risk scoring and dashboards
- Alert trend analysis
- Incident timelines & correlation
- Continuous risk scoring
- Custom organizational intelligence
- SIEM integration